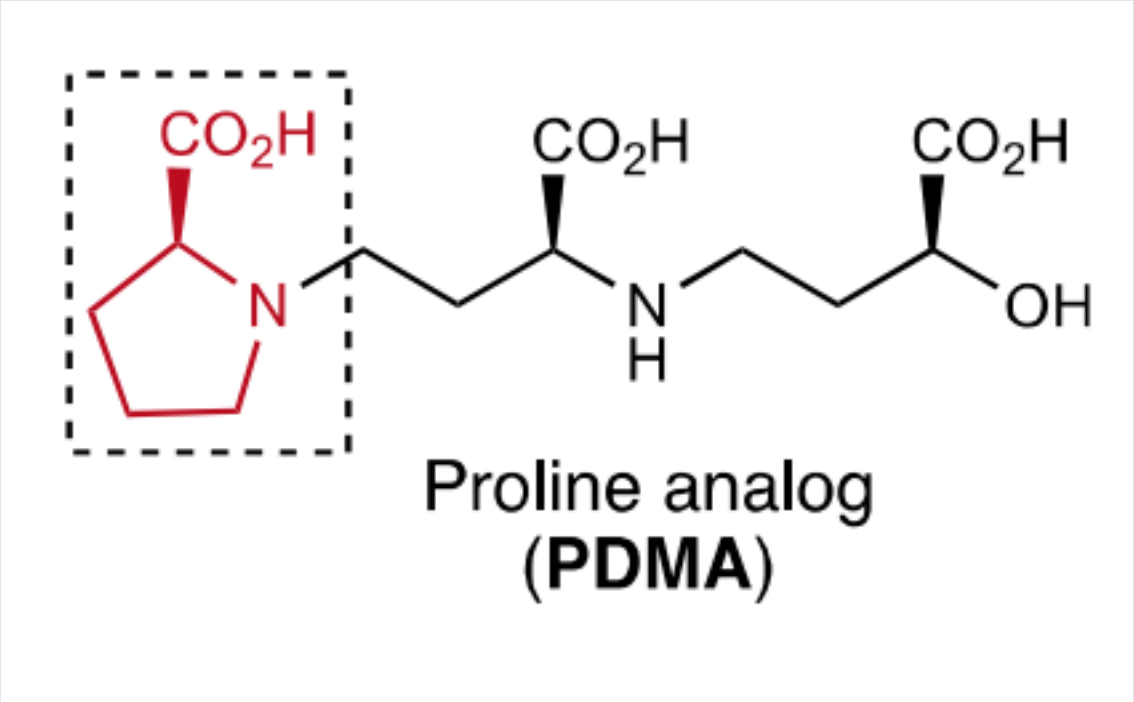
Achieving Internal Consistency in Databases used by Aqueous Modelling Software

BY: JOSHUA LEIPER

PRIMARY SUPERVISOR: DR. WEISS

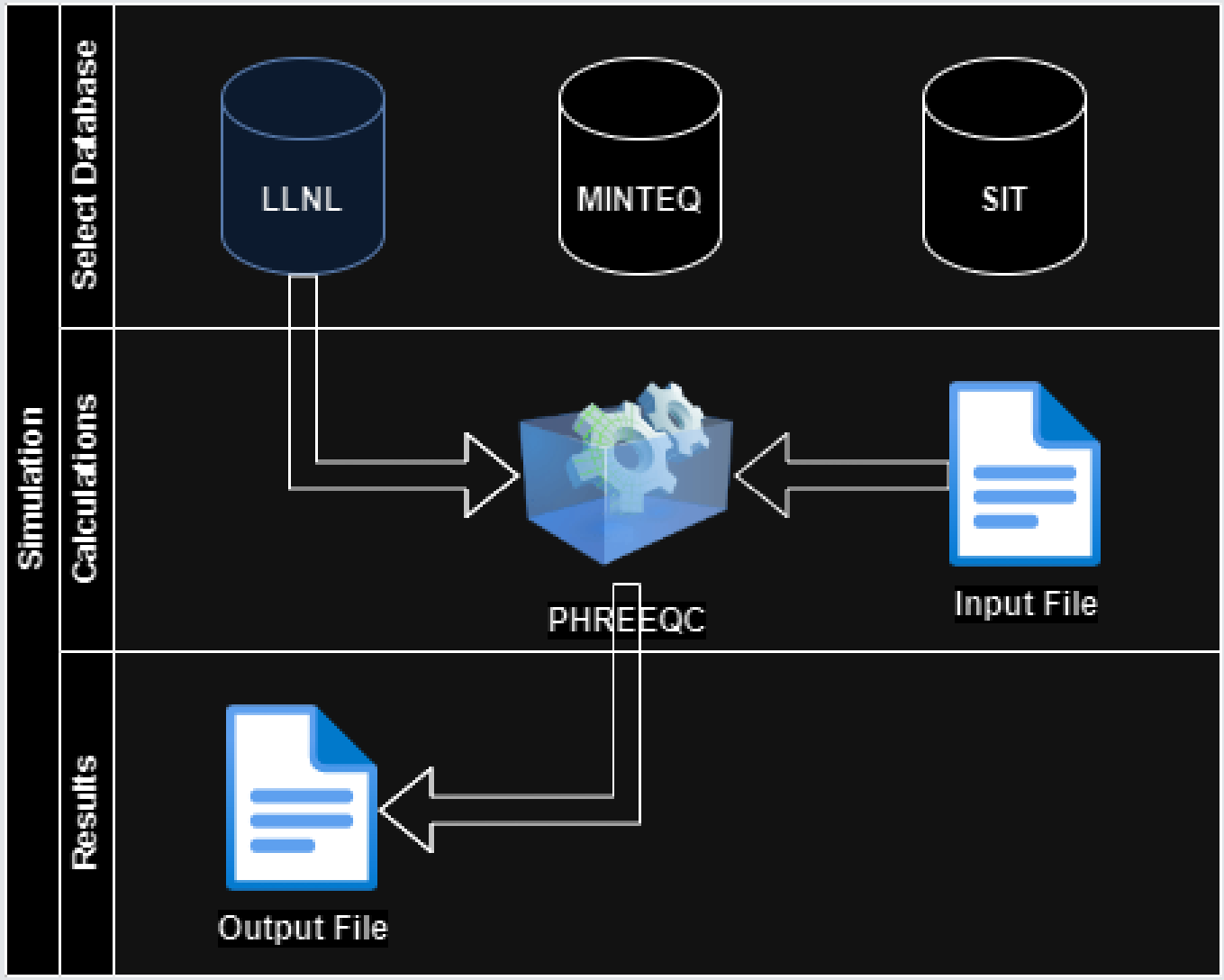
SECONDARY SUPERVISOR: DR. LIU



Motivation: Ligand-based fertilizer

PDMA is a molecule currently being researched as a possible fertilizer to increase rice yields. Geochemical software can be used to simulate a variety of environments that the fertilizer might be used in.

PHREEQC and the usage of geochemical simulation software



SOLUTION_MASTER_SPECIES & SOLUTION_SPECIES

```
SOLUTION_MASTER_SPECIES

#element      species       alk          gfw_formula    element_gfw
E             e-             0           0              0
Acetate       Acetate-       0           Acetate        59.01
Adipate       Adipate-2      0           Adipate        144.07
Ag            Ag+            0           Ag             107.8682
Al            Al+3           0           Al             26.9815
Alkalinity    CO3-2          1           Ca0.5(CO3)0.5  50.05
Am            Am+3           0           Am             243
Am(+2)        Am+2           0           Am             243
Am(+3)        Am+3           0           Am             243
Am(+4)        Am+4           0           Am             243
Am(+5)        AmO2+          0           Am             243
Am(+6)        AmO2+2         0           Am             243
As            AsO4-3         0           As             74.9216
As(+3)        H3(AsO3)       0           As             74.9216
```

```
SOLUTION_SPECIES

1.000H2O     = H2O
     log_k     0.000
     delta_h   0.000     #kJ/mol
     # Enthalpy of formation:
     -analytic 0E+0 0E+0 0E+0 0E+0 0E+0

1.000H+      = H+
     log_k     0.000
     delta_h   0.000     #kJ/mol
     # Enthalpy of formation:          0
     -analytic 0E+0 0E+0 0E+0 0E+0 0E+0

1.000e-      = e-
     log_k     0.000
     delta_h   0.000     #kJ/mol
     # Enthalpy of formation:          0       #kJ/mol       #89COX/WAG
     -analytic 0E+0 0E+0 0E+0 0E+0 0E+0
```

Objectives

1. Use Python to scrape PHREEQC database files
2. Compile SMS and SS into Pandas Dataframes
3. Inspect databases for conflicting values
4. Use analysis to chose baseline database and append unique information from other databases
5. Write database back into PHREEEQC database format

The problem – databases

Element	Species	Source
As	$H2AsO_4^-$	LLNL
As	H_3AsO_4	MINTEQ
As	AsO_4^{-3}	SIT

Databases are independently developed

Elements can be defined with different species

Similar information have different values

Each database has its own syntax and definition paradigm

SOLUTION_MASTER_SPECIES Conflicts

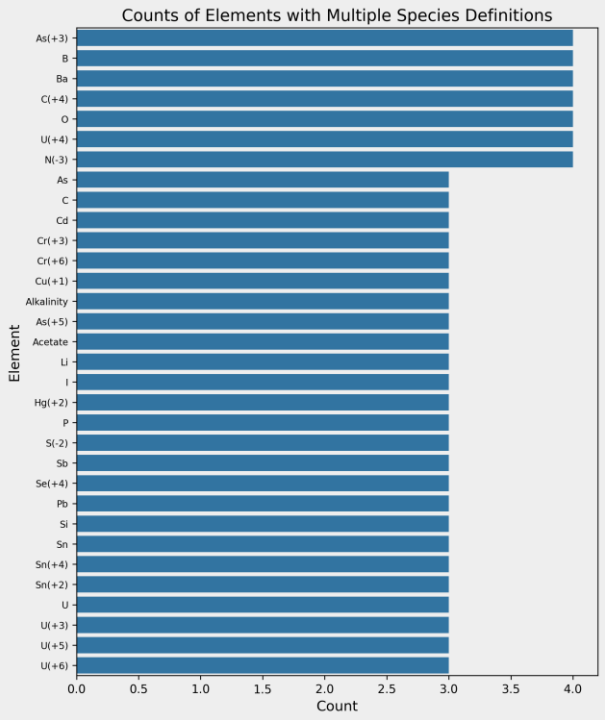
Alkalinity

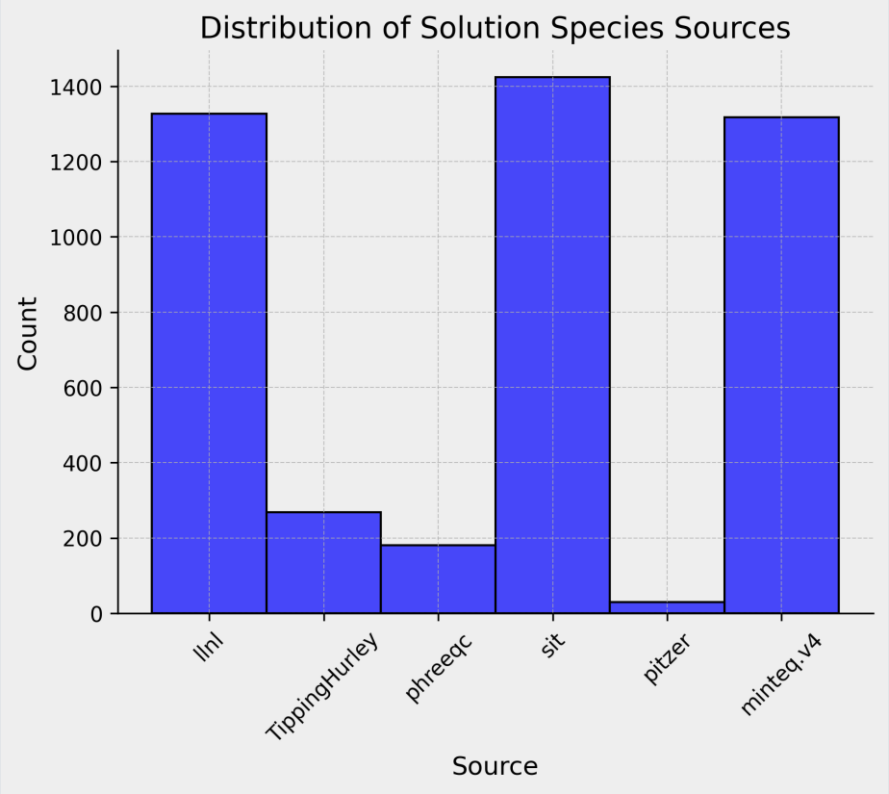
	element	species	alk	source
115	Acetate	Acetate-	1.0	#minteq.v4.dat
1	Acetate	Acetate-	0.0	#sit.dat
0	Alkalinity	CO3-2	2.0	#minteq.v4.dat
2	Alkalinity	CO3-2	1.0	#Tipping_Hurley.dat
25	Co(+3)	Co+3	-1.0	#minteq.v4.dat
53	Co(+3)	Co+3	0.0	#llnl.dat
26	Cr	CrO4-2	1.0	#minteq.v4.dat
54	Cr	CrO4-2	0.0	#llnl.dat
56	Cr(+3)	Cr+3	0.0	#llnl.dat
30	Cr(+3)	Cr+3	-1.0	#sit.dat
31	Cr(+6)	CrO4-2	1.0	#sit.dat
58	Cr(+6)	CrO4-2	0.0	#llnl.dat
36	Edta	Edta-4	0.0	#sit.dat
105	Edta	Edta-4	2.0	#minteq.v4.dat
43	Fe(+3)	Fe+3	0.0	#sit.dat

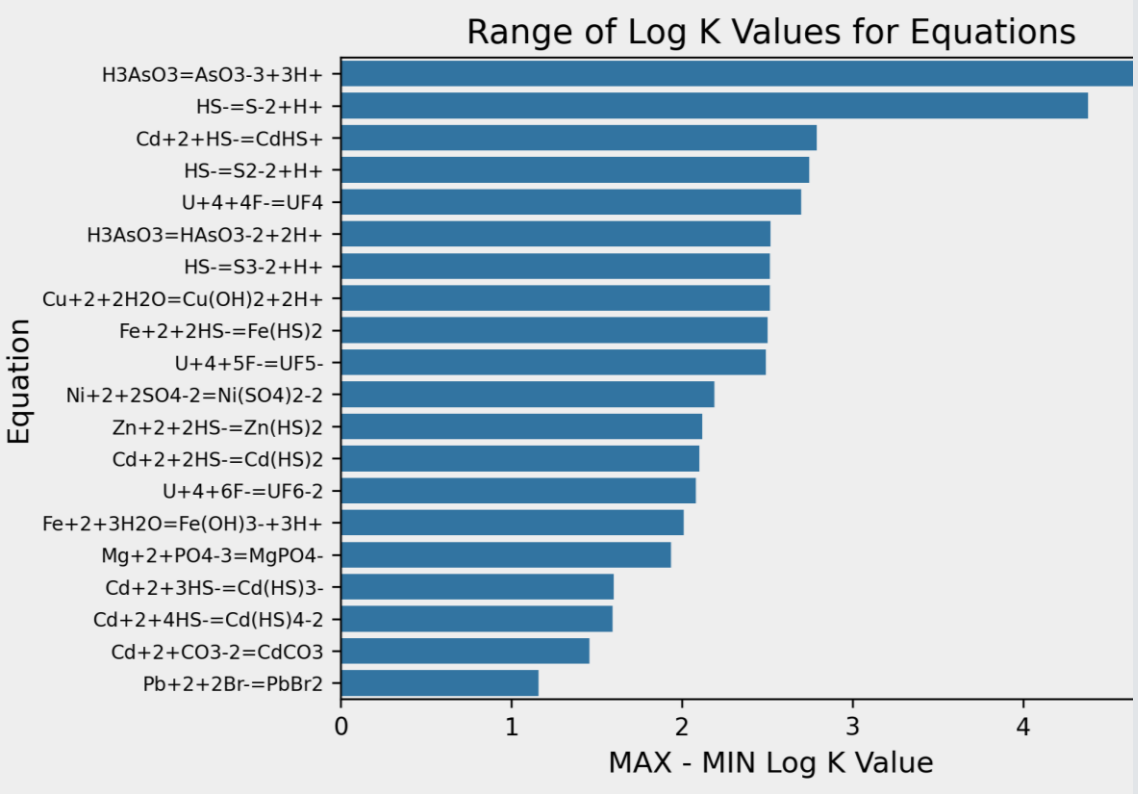
Gram Formula Weight

	element	species	element_gfw	source	element_gfw_range
4	Alkalinity	CO3-2	50.0500	#Tipping_Hurley.dat	10.9673
5	Alkalinity	CO3-2	61.0173	#minteq.v4.dat	10.9673
0	Acetate	Acetate-	59.0450	#minteq.v4.dat	0.0350
1	Acetate	Acetate-	59.0100	#sit.dat	0.0350
47	Ni	Ni+2	58.7100	#Tipping_Hurley.dat	0.0200
48	Ni	Ni+2	58.6900	#llnl.dat	0.0200
76	Zn	Zn+2	65.3900	#llnl.dat	0.0200
75	Zn	Zn+2	65.3700	#Tipping_Hurley.dat	0.0200
30	Cyanide	Cyanide-	26.0000	#llnl.dat	0.0177
29	Cyanide	Cyanide-	26.0177	#minteq.v4.dat	0.0177
10	Ba	Ba+2	137.3270	#sit.dat	0.0130
11	Ba	Ba+2	137.3270	#llnl.dat	0.0130
12	Ba	Ba+2	137.3300	#pitzer.dat	0.0130
13	Ba	Ba+2	137.3400	#Tipping_Hurley.dat	0.0130
21	Cd	Cd+2	112.4110	#llnl.dat	0.0110

Element-Species Definitions

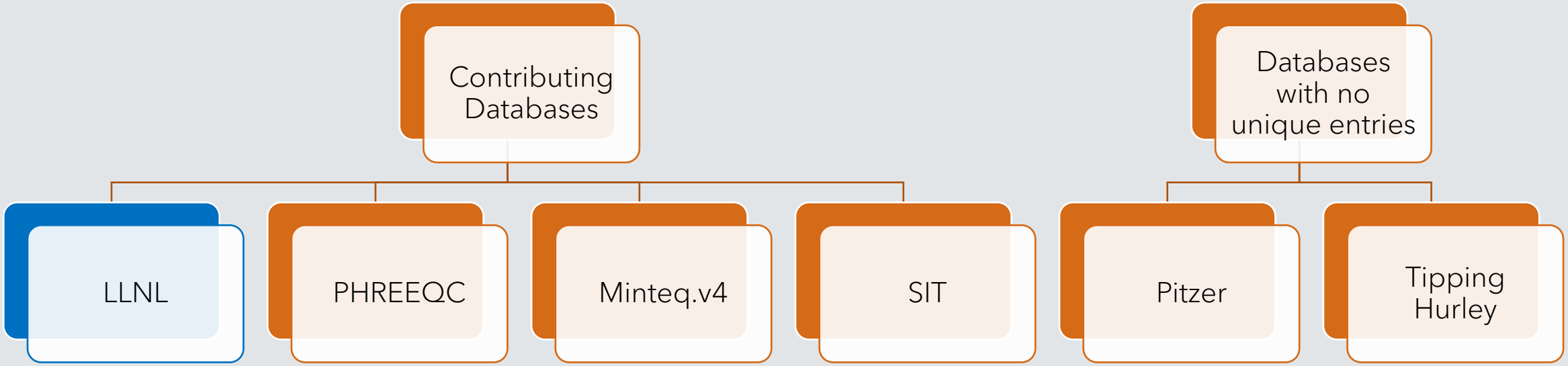






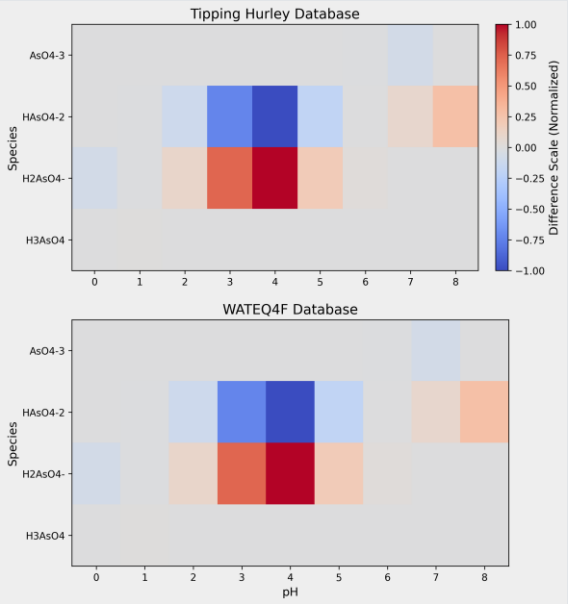
SOLUTION_SPECIES Conflicts

Unified Database composition

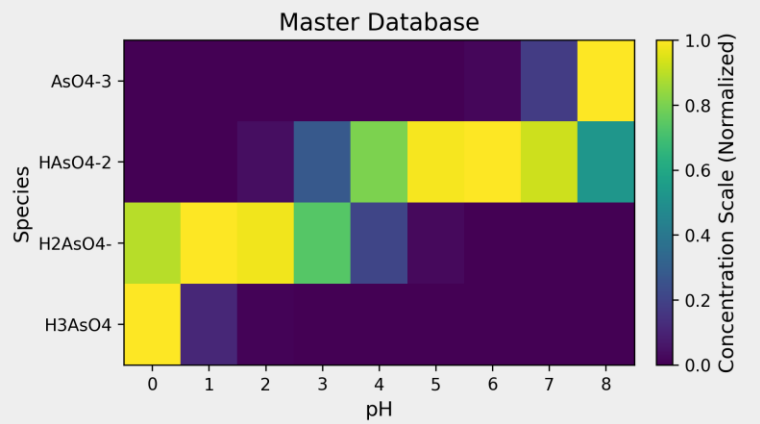


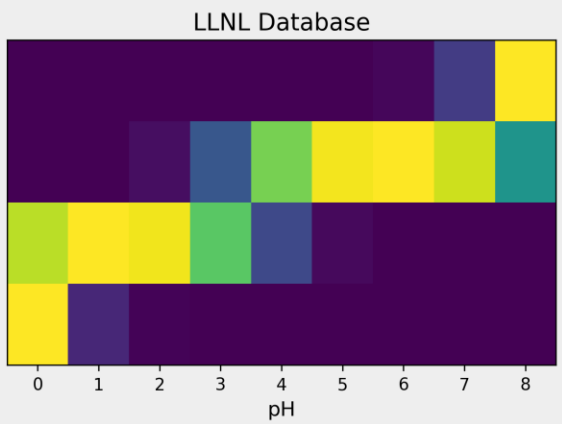
Database validation

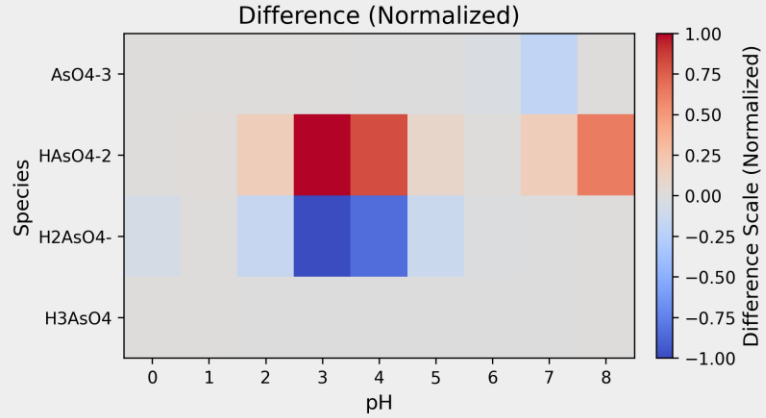
Same simulation with Tipping and WaterQ4F



Basic Simulation Using Master and LLNL databases







AsO4-3	HAsO4-2	H2AsO4-	H3AsO4
0.0	0.0	-0.000595	0.0
0.0	0.000167	0.0	7.2e-05
0.0	0.001723	-0.001483	-1e-05
0.0	0.009886	-0.009828	-1e-05
-0.0	0.008102	-0.00839	-1e-06
-2.1e-05	0.000932	-0.001279	-0.0
-0.000232	0.0	-0.000132	-0.0
-0.001822	0.001718	-1.1e-05	-0.0
0.0	0.006247	0.0	0.0

IMPERIAL

Future Work

- More validation trials of novel database
- Include other data blocks present in PHREEQC databases
- Develop more robust scraping methods for databases

Conclusion

- A comparative analysis of PHREEQC databases was conducted and areas of further investigation were identified
- A general-use database containing information from several PHREEQC databases was developed.

Citations

- Suzuki, Motofumi, et al. “Development of a Mugineic Acid Family Phytosiderophore Analog as an Iron Fertilizer.” *Nature Communications*, vol. 12, no. 1, 10 Mar. 2021, p. 1558, www.nature.com/articles/s41467-021-21837-6, https://doi.org/10.1038/s41467-021-21837-6. Accessed 25 Apr. 2022.